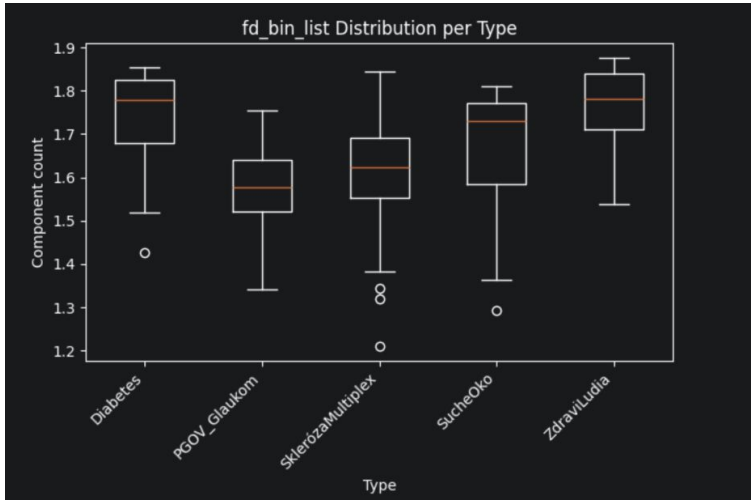
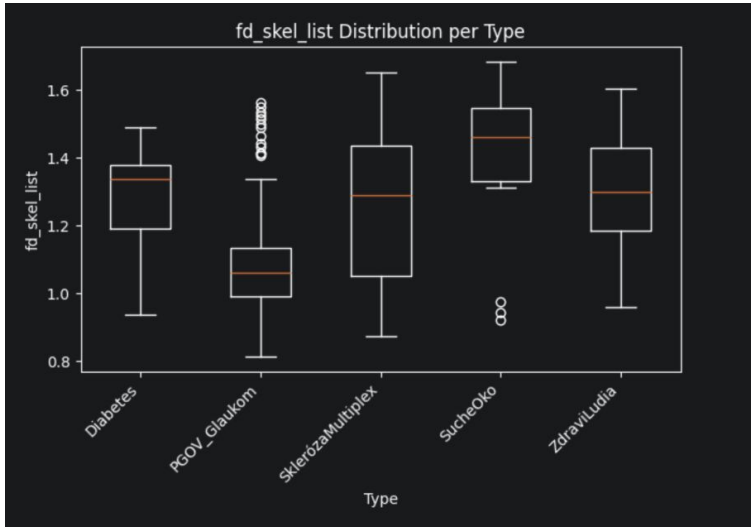
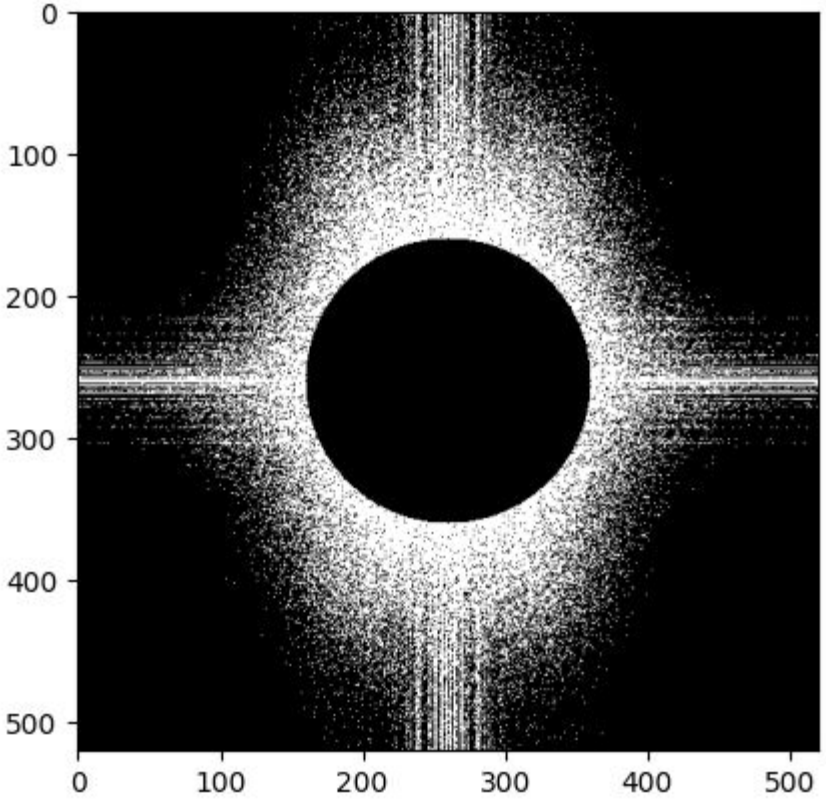
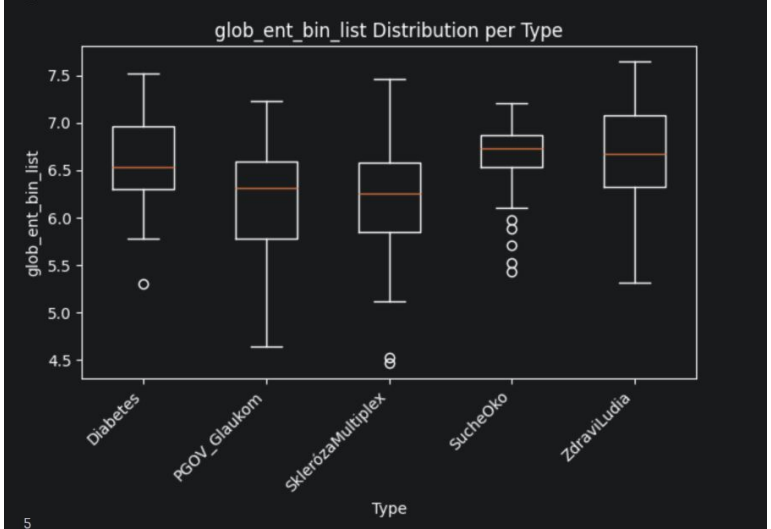
UPJŠ Challenge Analyzing Tears

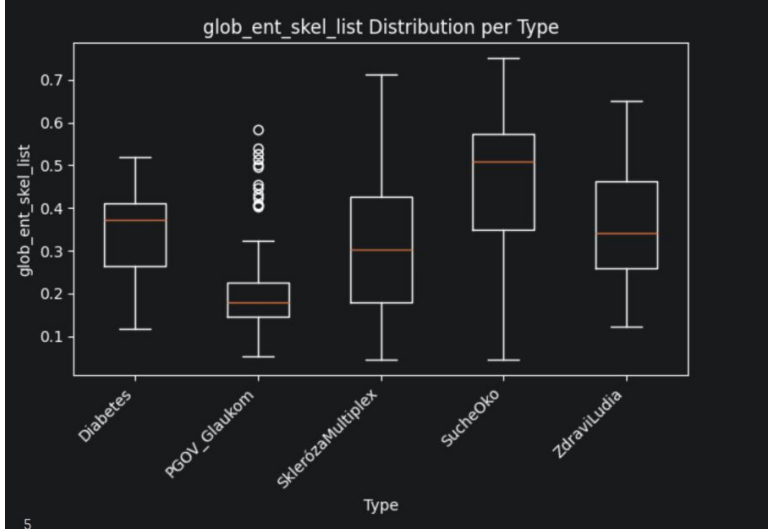
SUDO is bLoAt

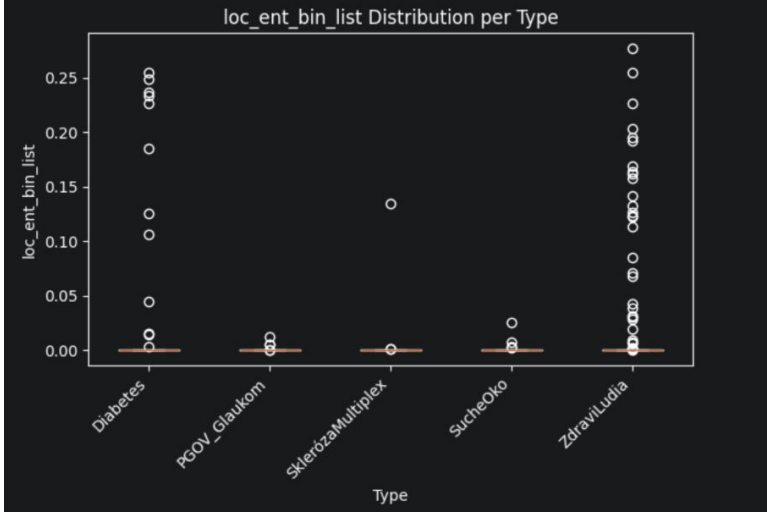

fd_bin_list Distribution per Type
Component count
1.9
1.8
1.7
1.6
1.5
1.4
1.3
1.2
Diabetes
PGOV_Glaukom
SklerózaMultiplex
SucheOko
ZdraviLudia
Type

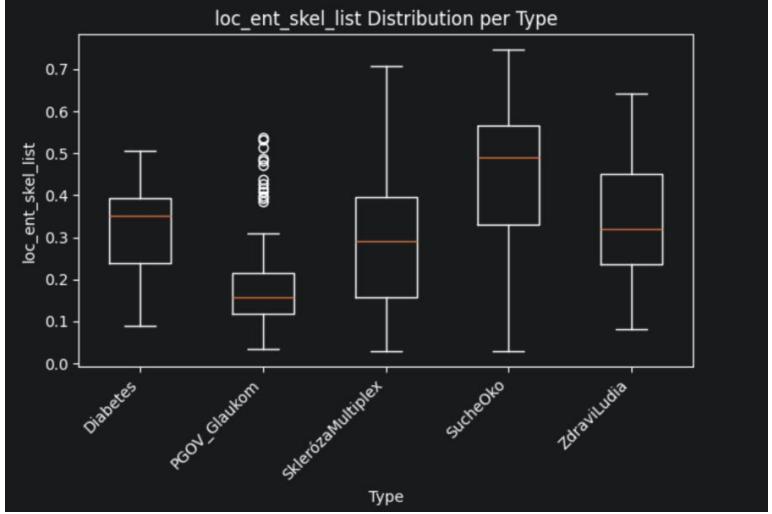

fd_skel_list Distribution per Type
fd_skel_list
1.6
1.4
1.2
1.0
0.8
Diabetes
PGOV_Glaukom
SklerózaMultiplex
SucheOko
ZdraviLudia
Type


0
100
200
300
400
500
0
100
200
300
400
500

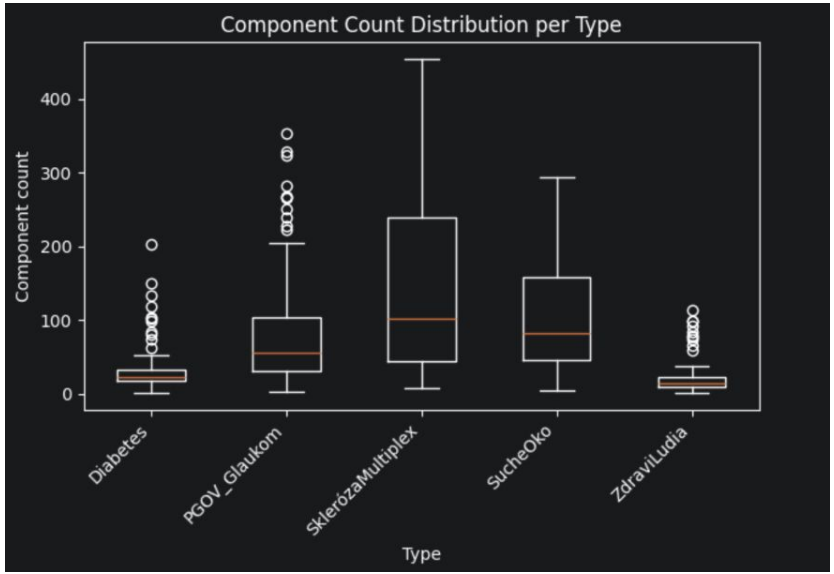

glob_ent_bin_list Distribution per Type
glob_ent_bin_list
7.5
7.0
6.5
6.0
5.5
5.0
4.5
Diabetes
PGOV_Glaukom
SklerózaMultiplex
SucheOko
ZdraviLudia
Type

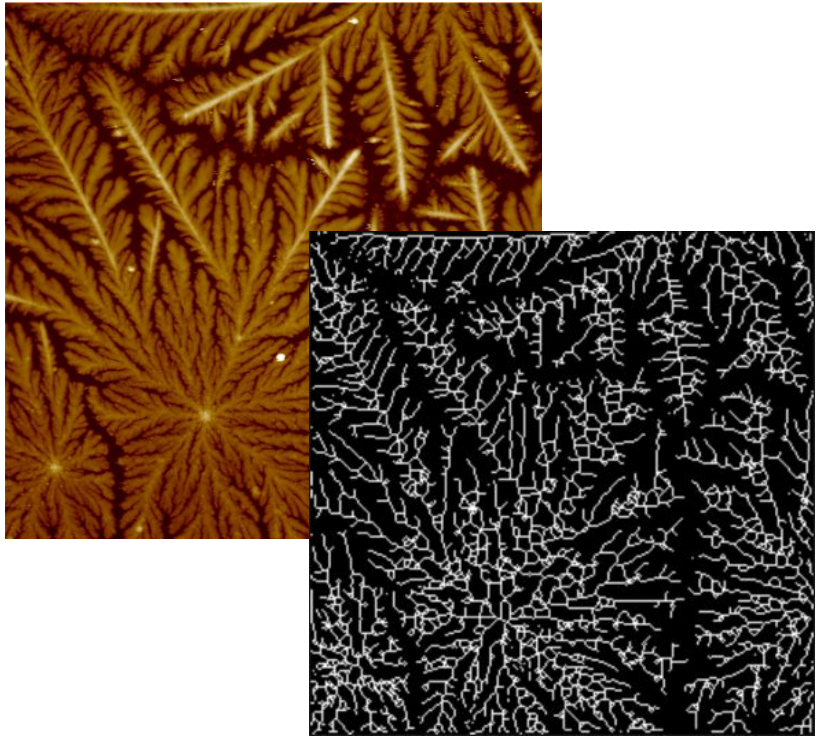

glob_ent_skel_list Distribution per Type
glob_ent_skel_list
0.7
0.6
0.5
0.4
0.3
0.2
0.1
Diabetes
PGOV_Glaukom
SklerózaMultiplex
SucheOko
ZdraviLudia
Type


loc_ent_bin_list Distribution per Type
loc_ent_bin_list
0.25
0.20
0.15
0.10
0.05
0.00
Diabetes
PGOV_Glaukom
SklerózaMultiplex
SucheOko
ZdraviLudia
Type

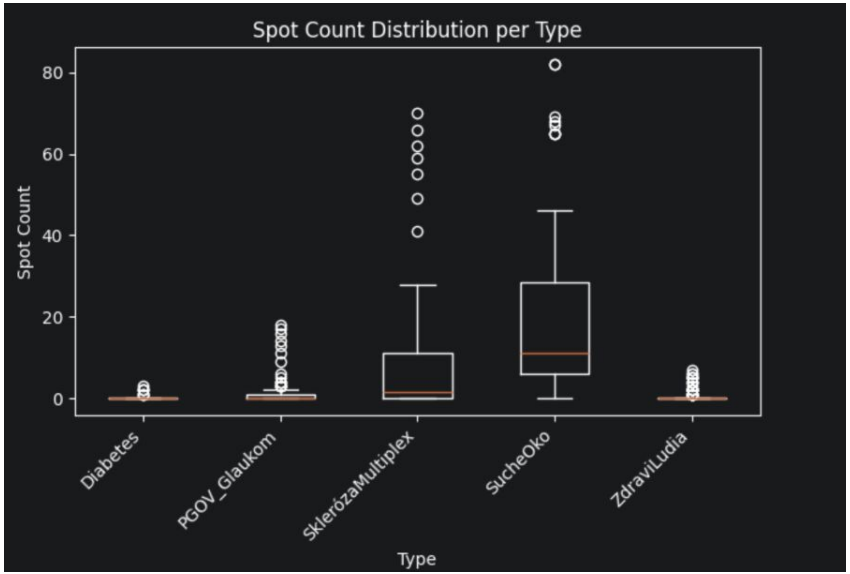

loc_ent_skel_list Distribution per Type
loc_ent_skel_list
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Diabetes
PGOV_Glaukom
SklerózaMultiplex
SucheOko
ZdraviLudia
Type

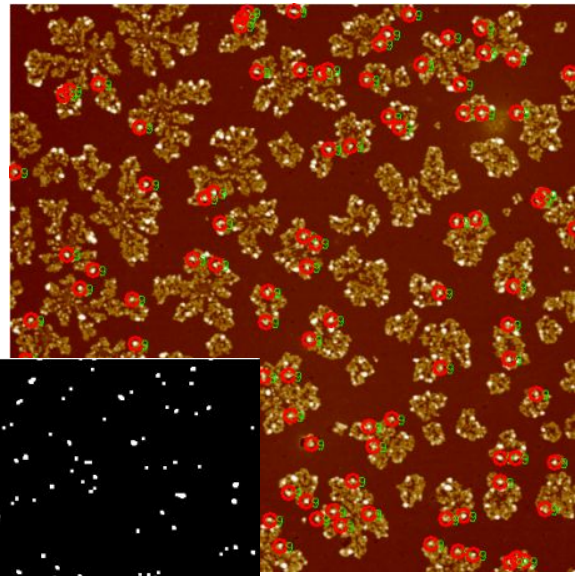
Our custom metrics - component count

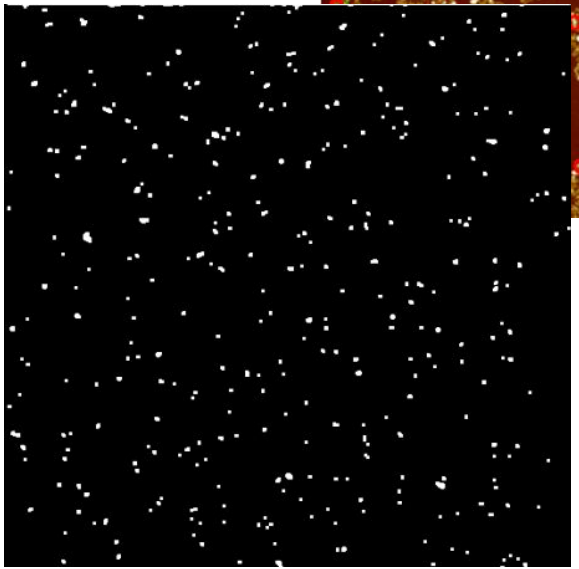




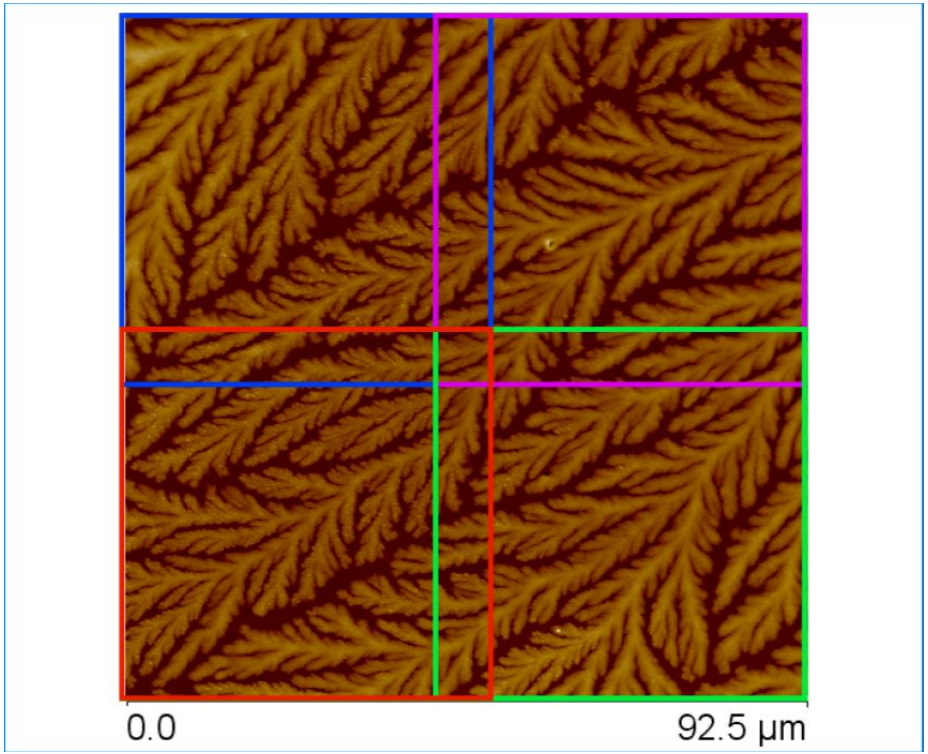
Our custom metrics - spot count

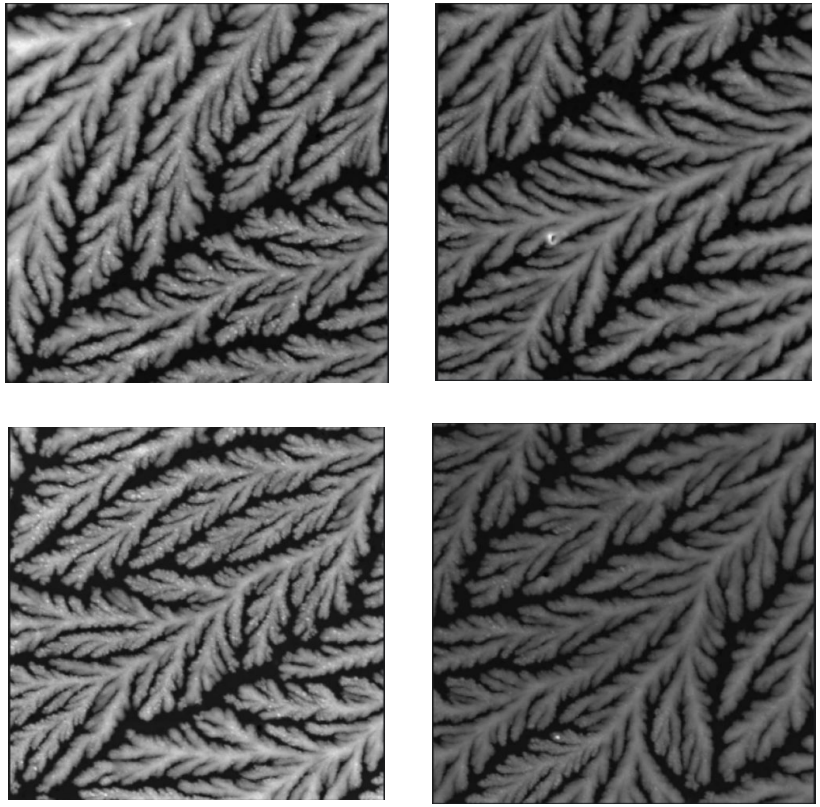




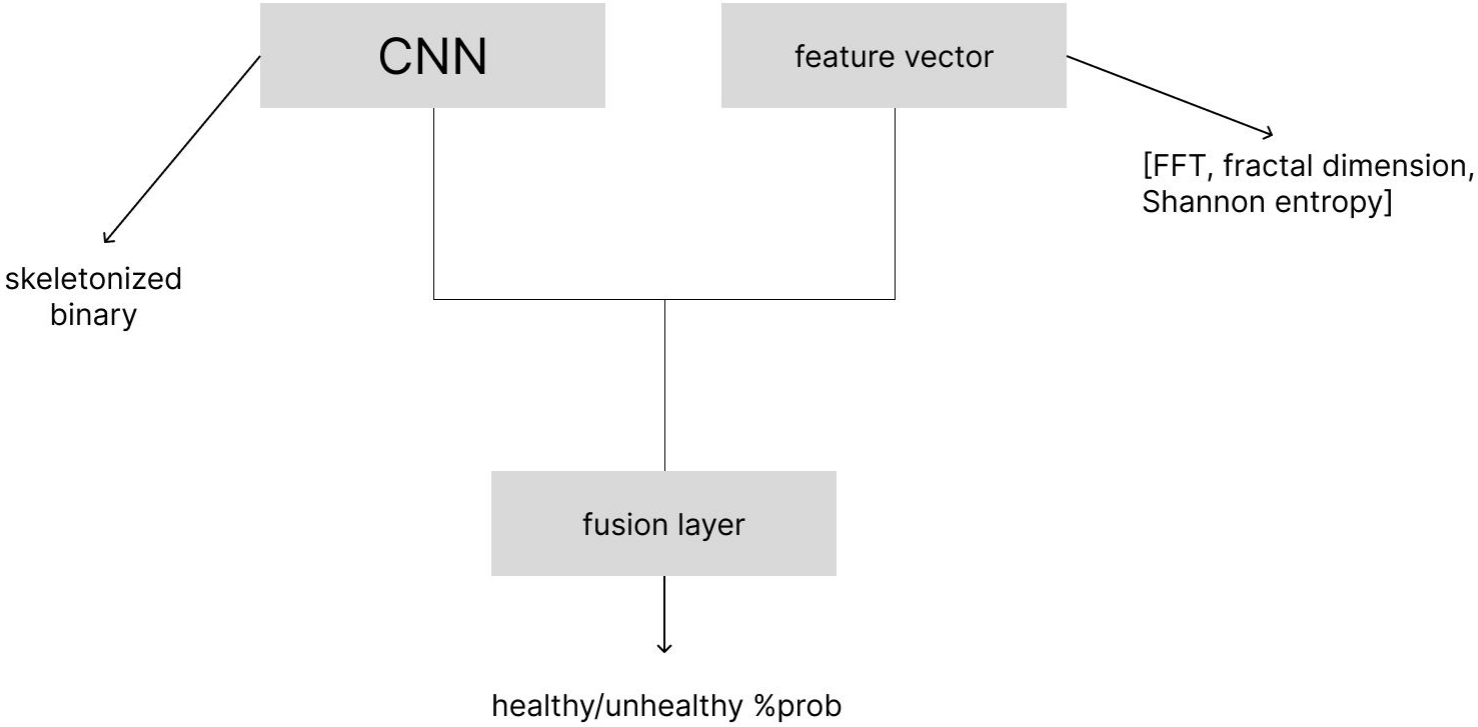


Being resourceful

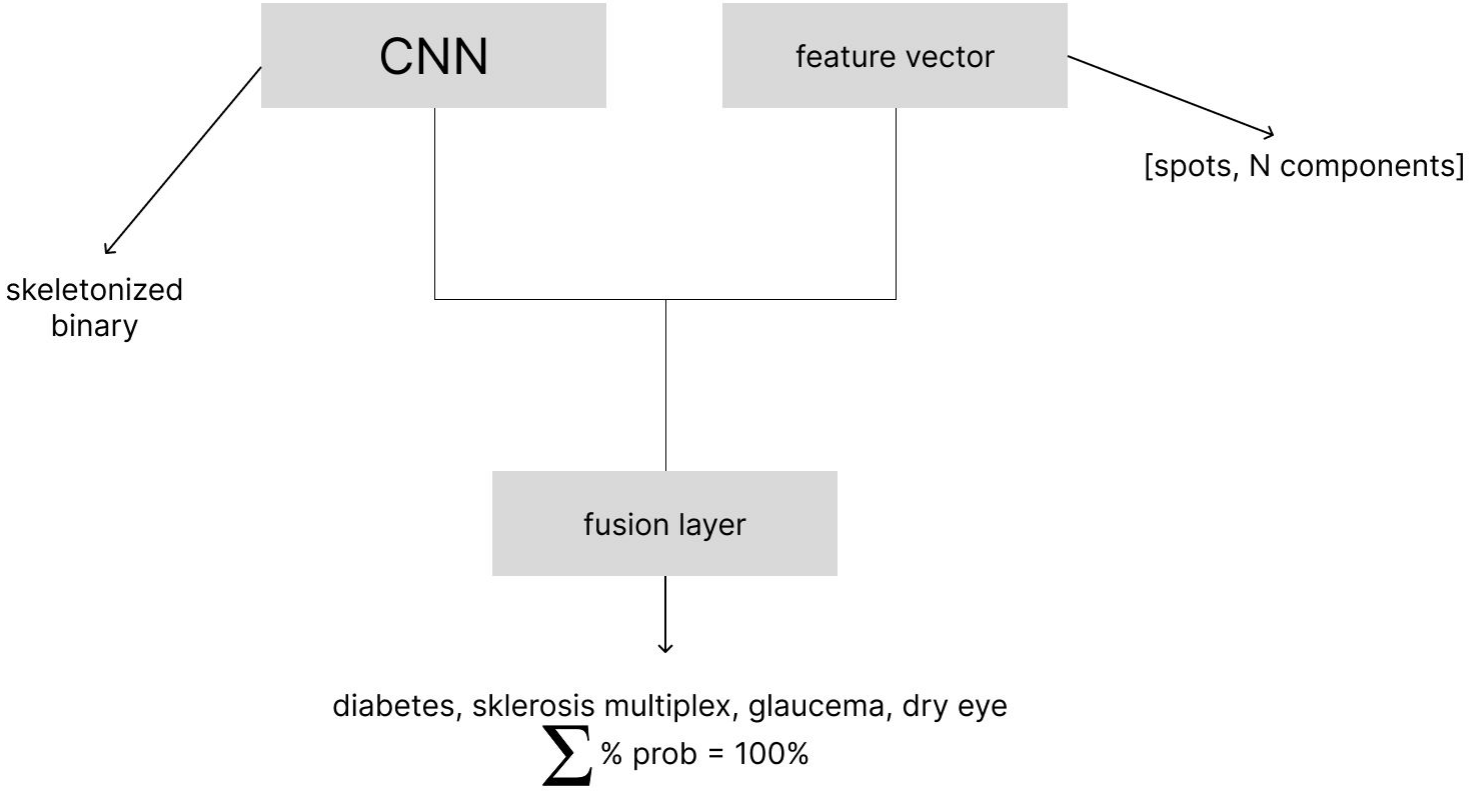




STAGE 1 MODEL ARCHITECTURE



STAGE 2 MODEL ARCHITECTURE

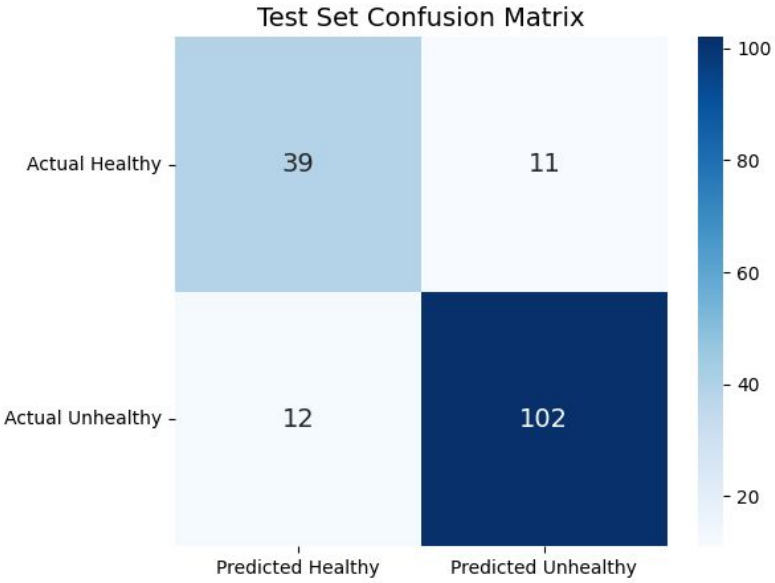


```
==================================================
CLASSIFICATION REPORT
==================================================
              precision    recall  f1-score   support

  Healthy (0)       0.76      0.78      0.77        50
Unhealthy (1)       0.90      0.89      0.90       114

     accuracy                           0.86       164
    macro avg       0.83      0.84      0.84       164
 weighted avg       0.86      0.86      0.86       164


==================================================
CONFUSION MATRIX
==================================================
```

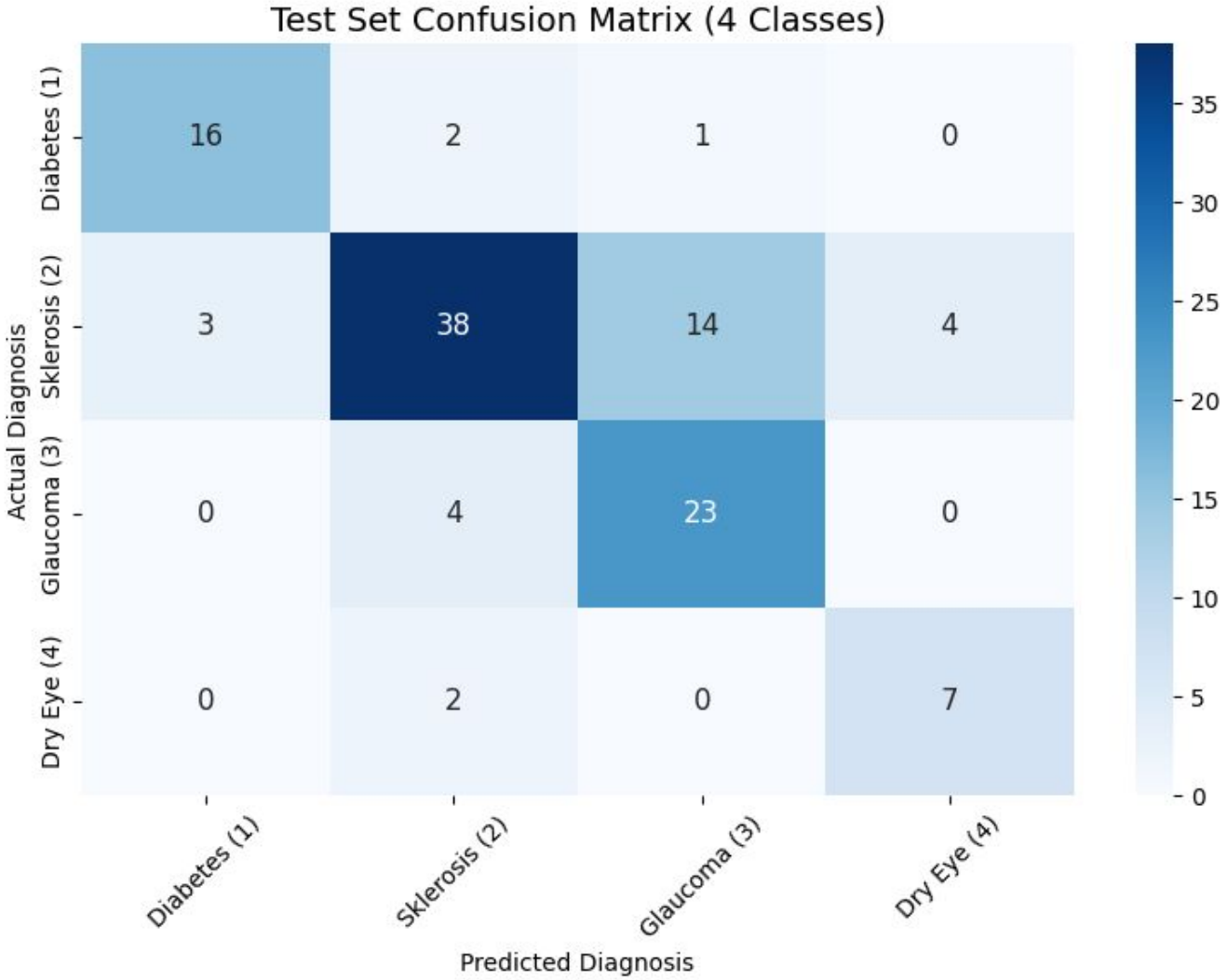


```
================================================================
MULTI-CLASS CLASSIFICATION REPORT
================================================================
               precision    recall  f1-score   support

 Diabetes (1)       0.84      0.84      0.84        19
Sklerosis (2)       0.83      0.64      0.72        59
 Glaucoma (3)       0.61      0.85      0.71        27
  Dry Eye (4)       0.64      0.78      0.70         9

     accuracy                           0.74       114
    macro avg       0.73      0.78      0.74       114
 weighted avg       0.76      0.74      0.74       114


================================================================
MULTI-CLASS CONFUSION MATRIX
================================================================
```


Test Set Confusion Matrix (4 Classes)
Actual Diagnosis
Diabetes (1)
Sklerosis (2)
Glaucoma (3)
Dry Eye (4)
16
2
1
0
3
38
14
4
0
4
23
0
0
2
0
7
Diabetes (1)
Sklerosis (2)
Glaucoma (3)
Dry Eye (4)
Predicted Diagnosis
35
30
25
20
15
10
5
0